

U-VLM: Hierarchical Vision Language Modeling for Report Generation

Pengcheng Shi¹, Minghui Zhang², Kehan Song¹, Jiaqi Liu¹, Yun Gu², Xinglin Zhang¹
¹Medical Image Insights Co. Ltd., Shanghai, China
²Shanghai Jiao Tong University, Shanghai, China
geron762@sjtu.edu.cn, xinglin.zhang@imagecore.com.cn

Motivation

Why is 3D report generation challenging: the model must perceive **multi-scale information**— from global anatomy down to lesion-level details (location, size, morphology)

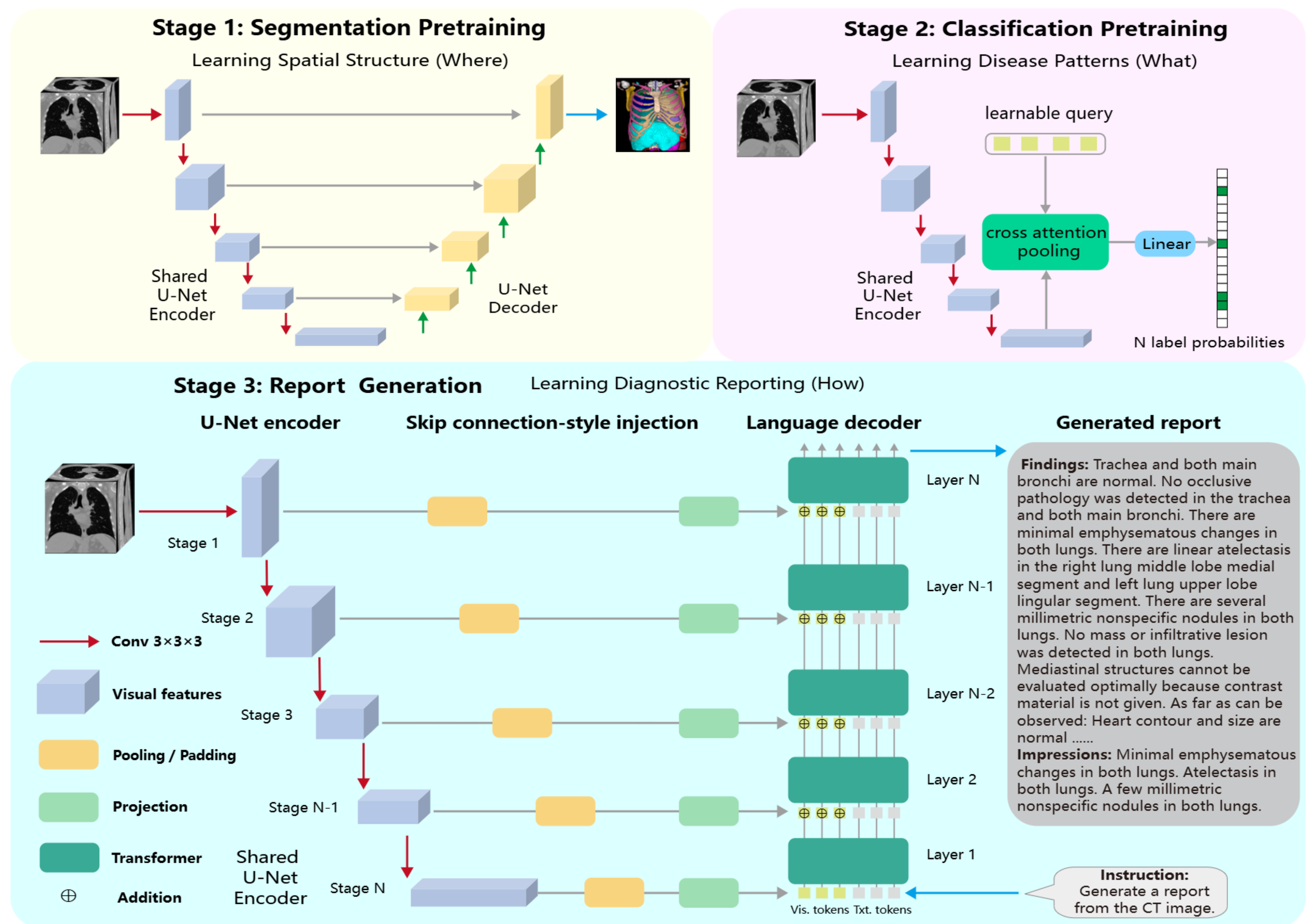
Yet existing 3D VLMs fall short:

- (1) **skip segmentation pretraining**, despite U-Net / nnU-Net dominating 3D segmentation
- (2) inject vision **only at the LLM input layer** → multi-scale detail is lost

Method	VE	LD	VE Init	LD Init	E2E
CT2Rep [9]	Causal Trans.	Trans. Dec.	Scratch	Scratch	✓
RadFM [22]	ViT+Perc.	MedLLaMA-13B	Scratch	Fine-tune	✓
M3D-LaMed [2]	ViT+Perc.	LLaMA-2-7B	CL	LoRA	✓
Merlin [5]	3D ResNet	RadLlama-7B	CL+Cls	LoRA	✓
CT-CHAT [11]	ViT+AP	LLaMA-3.1-70B	CL	LoRA	✓
MedM-VL [19]	ViT+AP	Qwen2.5-3B	CL	Fine-tune	✓
BTB3D [10]	Causal Conv.	LLaMA-3.1-8B	VQ Recon	LoRA	✓
RadGPT [3]	U-Net Enc.	—	Seg	—	✗
U-VLM (Ours)	U-Net Enc.	0.1B	Seg+Cls	Scratch	✓

Comparison with related work on 3D radiology report generation. VE: Vision Encoder. LD: Language Decoder. E2E: End-to-End. VE Init: CL (Contrastive Learning, CLIP-style), VQ Recon (vector-quantized reconstruction), Cls (classification), Seg (segmentation). Perc.: Perceiver. AP: Attention Pooling.

Overview



Proposed Method

Progressive Training (curriculum: where → what → how)

Stage 1 · Segmentation — dense per-voxel supervision learns spatial structure:

$$\mathcal{L}_{seg} = \mathcal{L}_{dice}(\mathbf{Y}_{seg}, \hat{\mathbf{Y}}_{seg}) + \mathcal{L}_{ce}(\mathbf{Y}_{seg}, \hat{\mathbf{Y}}_{seg})$$

Stage 2 · Classification — learnable queries aggregate encoder features:

$$\hat{\mathbf{y}} = \sigma(\text{Linear}(\text{Flatten}(\text{CrossAttn}(\mathbf{Q}, \mathbf{f}_N))))), \quad \mathcal{L}_{cls} = \text{BCE}(\mathbf{y}, \hat{\mathbf{y}})$$

Stage 3 · Report Generation:

$$\mathcal{L}_{gen} = - \sum_{t=1}^T \log P(w_t | w_{<t}, \{\mathbf{h}_j\}_{j=1}^L)$$

Multi-Layer Visual Injection (U-Net skip connections → VLM)

Feature alignment pools shallow stages / pads deep stages to one token length K; deep encoder stages feed early LM layers, shallow stages feed later ones:

$$\tilde{\mathbf{f}}_i = \text{Align}_r(\mathbf{f}_i) = \begin{cases} \text{Pool}(\mathbf{f}_i) & \text{if } i < r \\ \mathbf{f}_i & \text{if } i = r \\ \text{Pad}(\mathbf{f}_i) & \text{if } i > r \end{cases}$$

$$\mathbf{h}_j^{(v)} = \text{LM}_j \left(\mathbf{h}_{j-1}^{(v)} + \text{Proj}_j(\text{Align}_r(\mathbf{f}_{N-j+1})) \right), \quad \mathbf{h}_j^{(t)} = \text{LM}_j \left(\mathbf{h}_{j-1}^{(t)} \right)$$

Notations: None: random initialization (no pretraining); Seg(C): coarse anatomy; Seg(C+L): coarse anatomy + lesions; Seg(F+L): fine-grained anatomy + lesions. B-M: BLEU-mean. SC: Skip Connections; Frz: Frozen Encoder; VT: Visual Token

Training	CT-RATE			AbdomenAtlas 3.0		
	F1	Precision	Recall	F1	Precision	Recall
None → Cls	0.451	0.541	0.404	0.373	0.304	0.557
Seg(C) → Cls	0.546	0.587	0.535	0.617	0.568	0.679
Seg(C+L) → Cls	0.553	0.595	0.541	0.646	0.595	0.716
Seg(F+L) → Cls	0.555	0.598	0.539	0.578	0.567	0.589

Classification ablation on CT-RATE and AbdomenAtlas 3.0

Conclusion

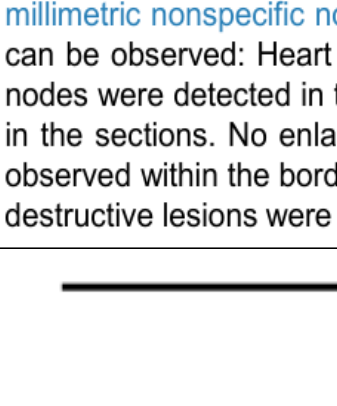
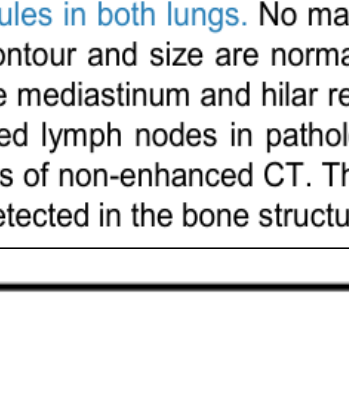
• Hierarchical VLM in **both training and architecture**: progressive pretraining significantly improves F1, while multi-layer injection improves BLEU-mean.

• For report generation, **vision encoder pretraining outweighs LLM scale**: seg-pretrained encoder + 0.1B scratch decoder beats 7B+ text-pretrained LLMs.

Results

Method	F1	Prec.	Rec.	B-1	B-2	B-3	B-4	B-mean
CT2Rep [9]	0.160	0.435	0.128	0.372	0.292	0.243	0.213	0.280
Merlin [5]	0.160	0.295	0.112	0.231	0.163	0.124	0.099	0.154
CT-CHAT [11]	0.184	0.450	0.158	0.373	0.284	0.231	0.198	0.272
MedM-VL [19]	0.167	0.323	0.160	0.390	0.295	0.240	0.205	0.283
BTB3D-8 [10]	0.187	0.260	0.150	0.411	0.307	0.245	0.215	0.295
BTB3D-16 [10]	0.258	0.260	0.260	0.420	0.318	0.256	0.225	0.305
U-VLM (Ours)	0.414	0.491	0.429	0.474	0.367	0.300	0.256	0.349

Report generation on CT-RATE

Report generation on chest CT (CT-RATE)		Ground Truth
Findings	Impressions	
 Findings: Trachea and both main bronchi are open. No occlusive pathology was detected in the trachea and both main bronchi. There are linear atelectasis in the right lung middle lobe and medial upper lobe anterior segment, and left lung upper lobe lingular segment. Minimal emphysematous changes were observed in both lungs. There is a millimetric nodule in the upper lobe of the right lung. No mass or infiltrative lesion was detected in both lungs. Mediastinal structures cannot be evaluated optimally because contrast material is not given. As far as can be observed: Heart contour and size are normal. No pleural or pericardial effusion was detected. The widths of the mediastinal main vascular structures are normal. No pathologically enlarged lymph nodes were detected in the mediastinum and hilar regions. No pathological wall thickness increase was observed in the esophagus within the sections. No upper abdominal free fluid-collection was detected in the sections. No enlarged lymph nodes in pathological dimensions were detected. In the upper abdominal organs within the sections, there is no mass with distinguishable borders as far as it can be observed within the borders of non-enhanced CT. Thoracic vertebral corpus heights, alignments and densities are normal. There are osteophytes in the vertebral corpus corners. Intervertebral disc distances are preserved. The neural foramina are open. Impressions: Minimal emphysematous changes in both lungs. Linear atelectasis in both lungs. Millimetric nodule in the upper lobe of the right lung. Minimal thoracic spondylitis.	 Impressions: Minimal emphysematous changes in both lungs. There are linear atelectasis in the right lung middle lobe medial segment and left lung upper lobe lingular segment. There are several millimetric nonspecific nodules in both lungs. No mass or infiltrative lesion was detected in both lungs. Mediastinal structures cannot be evaluated optimally because contrast material is not given. As far as can be observed: Heart contour and size are normal. No pleural or pericardial effusion was detected. The widths of the mediastinal main vascular structures are normal. No pathologically enlarged lymph nodes were detected in the mediastinum and hilar regions. No pathological wall thickness increase was observed in the esophagus within the sections. No upper abdominal free fluid-collection was detected in the sections. No enlarged lymph nodes in pathological dimensions were detected. In the upper abdominal organs within the sections, there is no mass with distinguishable borders as far as it can be observed within the borders of non-enhanced CT. Thoracic vertebral corpus heights, alignments and densities are normal. Intervertebral disc distances are preserved. The neural foramina are open. No lytic-destructive lesions were detected in the bone structures within the sections. Impressions: Minimal emphysematous changes in both lungs. Atelectasis in both lungs. A few millimetric nonspecific nodules in both lungs.	U-VLM (Ours)
 Findings: Trachea and both main bronchi are open. No occlusive pathology was detected in the trachea and both main bronchi. There are minimal emphysematous changes in both lungs. There are linear atelectasis in the right lung middle lobe medial segment and left lung upper lobe lingular segment. There are several millimetric nonspecific nodules in both lungs. No mass or infiltrative lesion was detected in both lungs. Mediastinal structures cannot be evaluated optimally because contrast material is not given. As far as can be observed: Heart contour and size are normal. No pleural or pericardial effusion was detected. The widths of the mediastinal main vascular structures are normal. No pathologically enlarged lymph nodes were detected in the mediastinum and hilar regions. No pathological wall thickness increase was observed in the esophagus within the sections. No upper abdominal free fluid-collection was detected in the sections. No enlarged lymph nodes in pathological dimensions were detected. In the upper abdominal organs within the sections, there is no mass with distinguishable borders as far as it can be observed within the borders of non-enhanced CT. Thoracic vertebral corpus heights, alignments and densities are normal. Intervertebral disc distances are preserved. The neural foramina are open. No lytic-destructive lesions were detected in the bone structures within the sections. Impressions: Minimal emphysematous changes in both lungs. Millimetric nodules in both lungs.	 Impressions: Minimal emphysematous changes in both lungs. There are linear atelectasis in the right lung middle lobe medial segment and left lung upper lobe lingular segment. There are several millimetric nonspecific nodules in both lungs. No mass or infiltrative lesion was detected in both lungs. Mediastinal structures cannot be evaluated optimally because contrast material is not given. As far as can be observed: Heart contour and size are normal. No pleural or pericardial effusion was detected. The widths of the mediastinal main vascular structures are normal. No pathologically enlarged lymph nodes were detected in the mediastinum and hilar regions. No pathological wall thickness increase was observed in the esophagus within the sections. No upper abdominal free fluid-collection was detected in the sections. No enlarged lymph nodes in pathological dimensions were detected. In the upper abdominal organs within the sections, there is no mass with distinguishable borders as far as it can be observed within the borders of non-enhanced CT. Thoracic vertebral corpus heights, alignments and densities are normal. Intervertebral disc distances are preserved. The neural foramina are open. No lytic-destructive lesions were detected in the bone structures within the sections. Impressions: Minimal emphysematous changes in both lungs. Atelectasis in both lungs. A few millimetric nonspecific nodules in both lungs.	CT-CHAT

Training	SC	Frz	VT	Dec.	CT-RATE		Abdomen	
					F1	B-M	F1	B-M
None → Rep	✓	✓	384	0.1B	0.062	0.251	0.264	0.336
Cls → Rep	✓	✓	384	0.1B	0.292	0.294	0.384	0.391
Seg(C+L) → Rep	✓	✓	384	0.1B	0.232	0.297	0.348	0.341
Seg(F+L) → Rep	✓	✓	384	0.1B	0.238	0.278	0.398	0.392
Seg(C) → Cls → Rep	✓	✓	384	0.1B	0.399	0.298	0.592	0.423
Seg(C+L) → Cls → Rep	✓	✓	384	0.1B	0.409	0.279	0.620	0.413
Seg(F+L) → Cls → Rep	✓	✓	384	0.1B	0.415	0.303	0.560	0.407
Seg(C+L) → Cls → Rep	✓	✓	384	0.1B	0.407	0.298	0.624	0.437
Seg(F+L) → Cls → Rep	✓	✓	384	0.1B	0.414	0.349	0.553	0.417

Report generation ablation on CT-RATE and AbdomenAtlas 3.0

Acknowledgments

Supported by the National Key R&D Program (2022ZD012400), STCSM (25511102602), NSFC (62373243), and Shanghai Economy & Informatization Commission (AI for Science, 2025-GZL-RGZN-BTBX-02023).